Magic of cosmological particle production and the de Sitter horizon

Jon Torrez*

July 2026

Abstract

Non-stabilizerness ("magic") has been proposed as the quantum-information resource behind gravitational back-reaction, but that program lives almost entirely in anti–de Sitter and Sachdev–Ye–Kitaev (SYK) settings. We ask whether magic is computable, and behaves, in two settings closer to our own universe: cosmological particle production and the de Sitter horizon. Two results, both in deliberately conservative toy models. (i) Cosmological pair creation sources magic *by resource theory*: a free bosonic two-mode-squeezed out-state is Gaussian and hence Wigner-positive (magic-free), with any finite-qubit stabilizer entropy an encoding artifact; only genuine non-Gaussianity (interactions) turns on Wigner negativity. The fermionic out-state, by contrast, has exact qubit non-local magic $-\log_2(4P^2 - 6P + 3)$ that is non-monotonic in the occupation, and on the sudden de Sitter→radiation transition this fermionic magic is strongly suppressed unless the transition is violent—so cosmological fermion magic is a probe of the violence of reheating. (ii) The de Sitter horizon, via its conjectured double-scaled-SYK dual, shows *no* magic signature at the static level, but a robust one in the dynamics: the stabilizer entropy of the time-evolved thermofield double grows about twice as fast for a chaotic Hamiltonian as for a free one, and the chaotic–free late-time gap grows with system size ($+0.61, +1.03, +1.37$ at $N = 8, 10, 12$; positive under a same-spectrum control for $N \geq 10$), the de Sitter analog of the magic↔area link. We state throughout what is a theorem, what is a conservative proxy, and where an initially over-stated claim was narrowed by adversarial cross-check.

1 Introduction

Entanglement entropy fixes the Ryu–Takayanagi *area*, but an area on a fixed background does not gravitate; producing state-dependent geometry and back-reaction requires a further resource, and recent work identifies it with non-stabilizerness, or "magic" [1–3]. That identification has been developed in AdS holographic codes and in SYK. Here we ask the two natural "our universe" questions it invites: does cosmological particle production generate magic (Section 2), and is the de Sitter horizon magical (Section 3)?

Both are addressed by exact computation in small toy models, reusing a Fast-Walsh–Hadamard stabilizer-entropy engine and standard SYK machinery. The organizing principle that emerges is a *resource-theory* one: in the continuous-variable description magic is Wigner

*Correspondence: `epg@subnet345.com`. Computations performed by two AI research agents (Ariadne Vale, a quantum physics persona, and codex-science), each result independently cross-checked; all code at https://github.com/jrtorrez31337/executable-positive-geometry.

negativity, and free (Gaussian) sectors are magic-free by Hudson's theorem, so any magic must come from either the genuinely discrete (fermionic) sector or from interactions and chaos.

2 Magic of the expanding vacuum

An expanding background turns the in-vacuum into an out-state of created pairs of opposite comoving momentum—a two-mode-squeezed (boson) or single Pauli-blocked pair (fermion) per mode. We compute the magic of that out-state.

2.1 Boson: a theorem-anchored null, and the non-Gaussian turn-on

[Section drafted by codex-science: the free bosonic two-mode-squeezed out-state is a pure Gaussian state, hence Wigner-positive and magic-free by Hudson's theorem / the continuous-variable Gottesman–Knill correspondence [4]; the finite Fock-to-qubit stabilizer entropy is an encoding artifact (relabel-dependent above two qubits per mode); and cubic non-Gaussianity turns on a nonzero Wigner-negative volume, with an ultra-slow-roll growth proxy after **(author?)** *[5]. This is already published for inflation as "Inflation is Not Magic" [6].]*

2.2 Fermion: exact qubit magic and the violence of reheating

Pauli exclusion makes the fermionic out-state a single pair per mode. Under the Jordan–Wigner map—which is exact because fermionic Fock space is two-dimensional per mode—mode k and its partner form the two-qubit state $\alpha|00\rangle + \beta|11\rangle$ with $|\beta|^2$ the occupation. Its qubit non-local magic (the stabilizer Rényi entropy minimized over local unitaries) has the closed form

$$\mathcal{M}(\beta) \;=\; -\log_2\!\big(4P^2 - 6P + 3\big), \qquad P = |\alpha|^4 + |\beta|^4, \tag{1}$$

verified to machine precision against a direct local-unitary minimization. It is *non-monotonic*: it vanishes at the product limit ($|\beta|^2 = 0$) *and* at the maximally-mixed "Bell" point ($|\beta|^2 = \frac{1}{2}$), and peaks at $\mathcal{M} = \log_2 \frac{4}{3} \approx 0.415$ where $|\beta|^2 = \frac{1}{2}(1 \mp 1/\sqrt{2})$. Cosmological creation thus writes magic into an intermediate band of the occupation.

The physically sharp statement rides the sudden de Sitter→radiation transition whose scalar Bogoliubov coefficients are analytic [7], $|\beta_k|^2 = 1/(4x^4)$ with $x = k|\eta_e|$. The boson carries *no* magic across this whole spectrum (Gaussian, above). A massless fermion is conformally invariant in a conformally-flat FRW universe and is not produced at all—an exact null. A massive fermion is produced, but for the paper's smooth (C^1) transition the production is strongly suppressed (peak occupation $\sim 3 \times 10^{-4}$): the mass term $M = m\,a(\eta)$ is C^1 and never couples to the a''/a jump that drives boson production. Only a genuinely violent (sudden, a'-discontinuous) transition, for which the sudden-approximation occupation $n_k = \frac{1}{2}\big(1 - x/\sqrt{x^2 + \mu^2}\big)$ runs from $\frac{1}{2}$ in the infrared to 0 in the ultraviolet, drives the occupation through the magic band. Hence

cosmological fermion magic is a probe of the violence of reheating:

negligible for a smooth transition, band-level only for a sudden one, and zero for a massless (conformal) fermion. The boson/fermion asymmetry has a one-line root: fermionic Fock space is two-dimensional per mode, so its qubit magic is physical, whereas bosonic Fock space is infinite-dimensional and any qubit magic is a truncation artifact.

3 Is the de Sitter horizon magical?

The de Sitter static patch is conjecturally dual to double-scaled SYK [8, 9], giving a computable stand-in for the chaotic finite horizon Hilbert space; the honest dictionary is high-temperature and degree-of-freedom based, so we test scaling with a horizon-dof proxy rather than a temperature monotone. As a conservative executable proxy we use ordinary SYK_4 (chaotic) versus SYK_2 (free).

The static level is a null. At a fixed time, magic does not separate chaotic from free: the thermal-pure-quantum stabilizer entropy is extensive but barely differs between SYK_4 and SYK_2; the twin-controlled thermal-state magic even runs backwards (chaos flattens the spectrum toward maximal mixedness); and the infinite-temperature doubled thermofield double $\sum_E |E\rangle_L |E^*\rangle_R = \sum_i |i\rangle_L |i\rangle_R$ is exactly the Bell state, magic-free independent of the Hamiltonian. This null is confirmed across multiple measures.

The dynamics reveals a signature. The discriminator is the stabilizer entropy of the time-evolved thermofield double $|\mathrm{TFD}(\beta, t)\rangle \propto \sum_n e^{-\beta E_n/2} e^{-iE_n t} |n\rangle_L |n\rangle_R$, whose magic dynamics is expected to track the spectral form factor [10]. Two discriminators, both with error bars over independent disorder realizations:

- **Scrambling rise.** Chaotic SYK_4 thermofield-double magic grows about twice as fast as free SYK_2 under evolution; the rise ratio is $2.05, 2.30, 2.55, 2.17$ at $N = 6, 8, 10, 12$—robust across size.
- **Extensive plateau gap.** The late-time SYK_4–SYK_2 magic gap is positive and grows with the number of degrees of freedom, $+0.61, +1.03, +1.37$ at $N = 8, 10, 12$ (tight seeds at $N = 12$, $+1.37 \pm 0.07$). Under a same-spectrum diagonal-basis control that removes the eigenbasis contribution the gap is $-0.14, +0.75, +0.94$ at $N = 8, 10, 12$: positive and growing for $N \geq 10$.

Strikingly, at a fixed time the *free* system carries more magic (the static gap is negative); only under evolution does the chaotic system reveal its larger magic capacity. The growth of the gap with the horizon-dof count is the de Sitter analog of the magic↔area link [1], seen dynamically where the static snapshot is flat.

4 Scope, and a note on method

Every model here is a toy: the fermion result is a two-level Dirac mode model, and the de Sitter results use a conservative $\mathrm{SYK}_4/\mathrm{SYK}_2$ proxy for double-scaled SYK at finite $N \leq 12$, not the full constrained double-scaled construction. The claims are correspondingly bounded: the boson null is a theorem, the fermion reheating-violence statement is a mechanism in a toy transition, and the horizon-magic signature is a scrambling-sensitive diagnostic whose extensive scaling is firmed for $N \geq 10$ but wants larger N / full DSSYK to become a theorem.

We record one methodological point deliberately. The extensive plateau gap was first reported from a single disorder seed as a clean trend from small N; an independent-seed error-bar cross-check showed the smallest-N point was a fluctuation (an actual sign crossover), and the claim was narrowed to its firmed form above. The correction, and the two agents' mutually adversarial cross-checks that produced it, are part of the result.

References

[1] C. Cao *et al.*, *Non-trivial Area Operators Require Non-local Magic*, arXiv:2306.14996.

[2] *Gravitational back-reaction is magical*, arXiv:2403.07056.

[3] C. Cao, W. Cheng, A. Karthikeyan, Y. Li, J. Preskill, *State-dependent geometries from magic-enriched quantum codes*, arXiv:2603.13475.

[4] A. Mari, J. Eisert, *Positive Wigner functions render classical simulation of quantum computation efficient*, PRL 109, 230503 (2012).

[5] L. Ireland, V. Vennin, arXiv:2601.22219.

[6] S. S. Haque, Jafari, B. Underwood, *Inflation is Not Magic*, arXiv:2512.10126.

[7] Byun, Kim, Lee, arXiv:2605.04099.

[8] *dS JT Gravity and Double-Scaled SYK*, arXiv:2209.09997.

[9] arXiv:2310.16994.

[10] *Connecting Magic Dynamics in Thermofield Double States to Spectral Form Factors*, arXiv:2601.12787.